

CS 4782 Coding Assignment 5 Written Responses

Due: 5/5/25 11:59 PM on Gradescope

Late submissions accepted until 5/7/25 11:59 PM

Note: For homework, you can work in teams of up to 2 people. Please include your teammates' NetIDs and names on the front page and form a group on Gradescope. (Please answer all questions within each paragraph.) Please show all the relevant steps in your solutions.

Question 1:

How does the behavior of the Q-Learning agent compare to the random policy visualized previously? What policy does the agent appear to have learned? Write 2-3 sentences below. (5 pts)

Answer

Compared to random policy, it is obvious that the Q-learning agent exhibits more structured and goal-directed (the taxi driver do go to pick up passenger and drop off the passenger at the destination). However, the random policy selects actions uniformly at random, resulting in unpredictable trajectories that often fail to reach the goal or take inefficient paths.

Question 2:

How does the behavior of the DQN agent compare to the random policy visualized previously? What policy does the DQN agent appear to have learned? Write 2-3 sentences below. (5 pts)

Answer

The DQN agent learns how to balance the cartpole by applying forces in a controlled manner. From the visualization, the agent appears to have learned a policy that keeps the pole upright by continuously adjusting the cart's position. It moves the cart right to counteract the pole's tilt. However, the random policy applies force randomly, causing the pole to quickly lose balance and fall within a few steps.

Question 3:

Compare your results for using DQN and Reinforce. Discuss the performance, convergence, and behavior of the two algorithms. Consider factors such as the stability of learning, the quality of the learned policies, and the behavior of the final policy. Write 3-5 sentences below. (5 pts)

Answer

Compared with the DQN images, it is obvious that DQN converges faster than Reinforce, cause DQN achieves higher average rewards and learns more efficiently, while Reinforce shows higher variance and slower improvement. What's more, the policy learned by DQN is more deterministic and exploitative, while Reinforce learns a stochastic policy that maintains exploration even after training.

Gifs:

Make sure to upload all of the .gifs zipped with your written response. There should be 5 gifs total; two for the taxi (random, q-learning), three for the cartpole (random, dqn, reinforce).